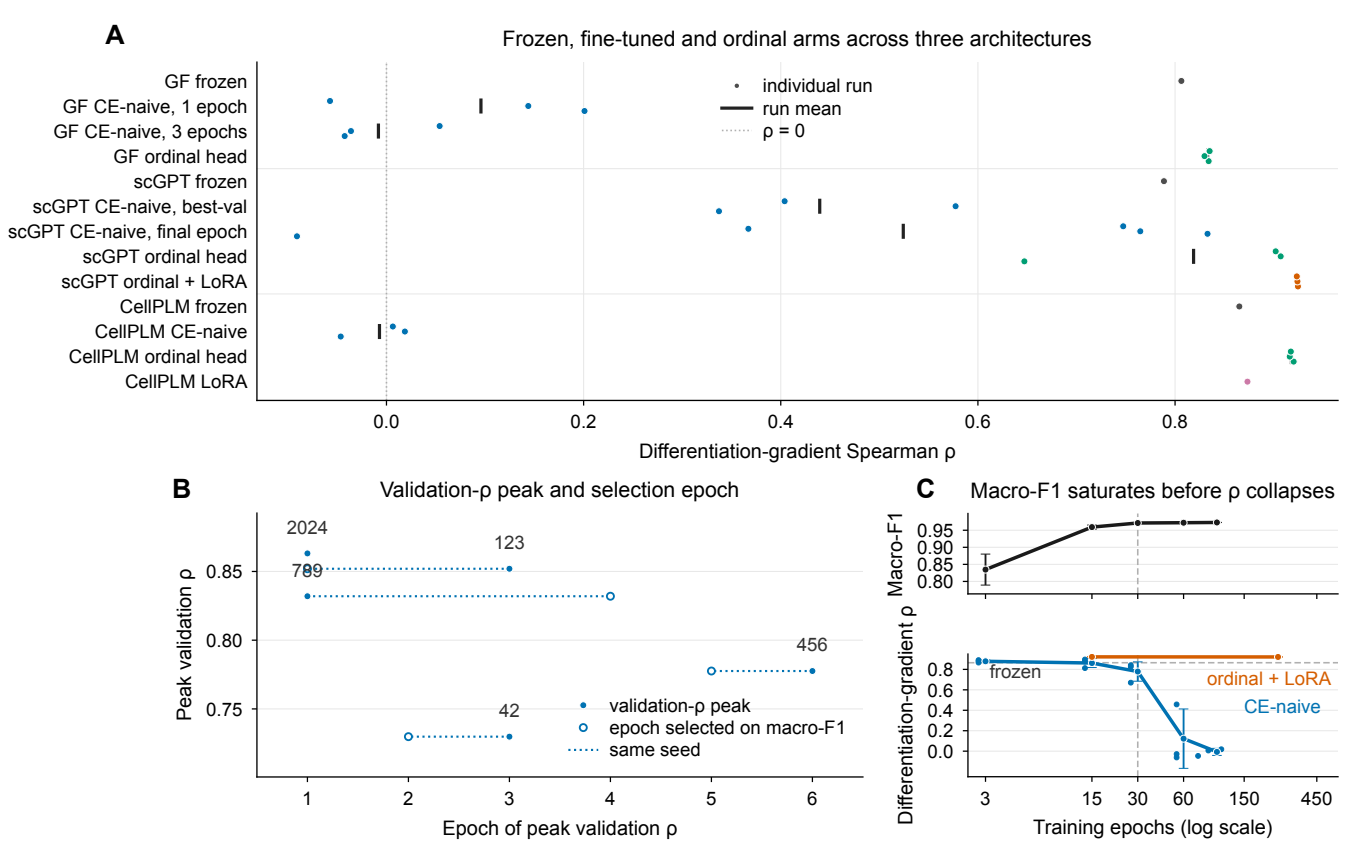
A
Frozen, fine-tuned and ordinal arms across three architectures
GF frozen
GF CE-naive, 1 epoch
GF CE-naive, 3 epochs
GF ordinal head
scGPT frozen
scGPT CE-naive, best-val
scGPT CE-naive, final epoch
scGPT ordinal head
scGPT ordinal + LoRA
CellPLM frozen
CellPLM CE-naive
CellPLM ordinal head
CellPLM LoRA
individual run
run mean
ρ = 0
0.0
0.2
0.4
0.6
0.8
Differentiation-gradient Spearman ρ
B
Validation-ρ peak and selection epoch
2024
123
789
456
42
Peak validation ρ
0.85
0.80
0.75
validation-ρ peak
epoch selected on macro-F1
same seed
1
2
3
4
5
6
Epoch of peak validation ρ
C
Macro-F1 saturates before ρ collapses
Macro-F1
0.95
0.90
0.85
0.80
Differentiation-gradient ρ
frozen
ordinal + LoRA
CE-naive
0.8
0.6
0.4
0.2
0.0
3
15
30
60
150
450
Training epochs (log scale)